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$$\int \frac{dx}{4x^2 + 9} = \int \frac{dx}{4 \left(x^2 + \frac{9}{4} \right)} = \int \frac{dx}{4 \left(x^2 + \left(\frac{3}{2} \right)^2 \right)} = \frac{1}{4} \int \frac{dx}{x^2 + \left(\frac{3}{2} \right)^2}$$

Formule is: $\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \operatorname{Bgtan} \frac{x}{a} + c$:

$$= \frac{1}{4} \cdot \frac{1}{\frac{3}{2}} \operatorname{Bgtan} \frac{x}{\frac{3}{2}} + c$$

$$= \frac{1}{6} \operatorname{Bgtan} \frac{2x}{3} + c$$

References